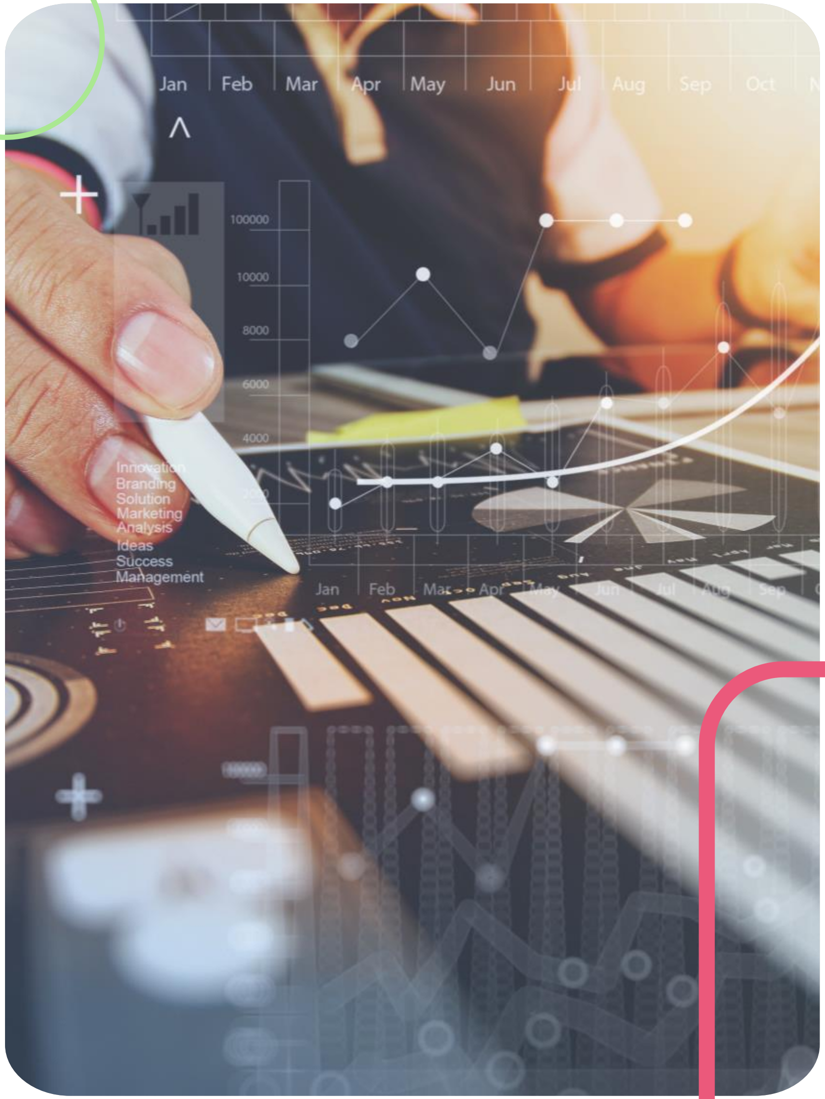
Diploma in

Data Analytics

Lesson 4: Linear regression continued...



Lesson Objectives

Lesson 4

- Linear regression continued
- Data frames
- Dates in R

Linear regression continued

Recap linear regression

- $\hat{y} = a + bx_i$
- lm.fit()
- Summary(lm.fit)

Multiple linear regression

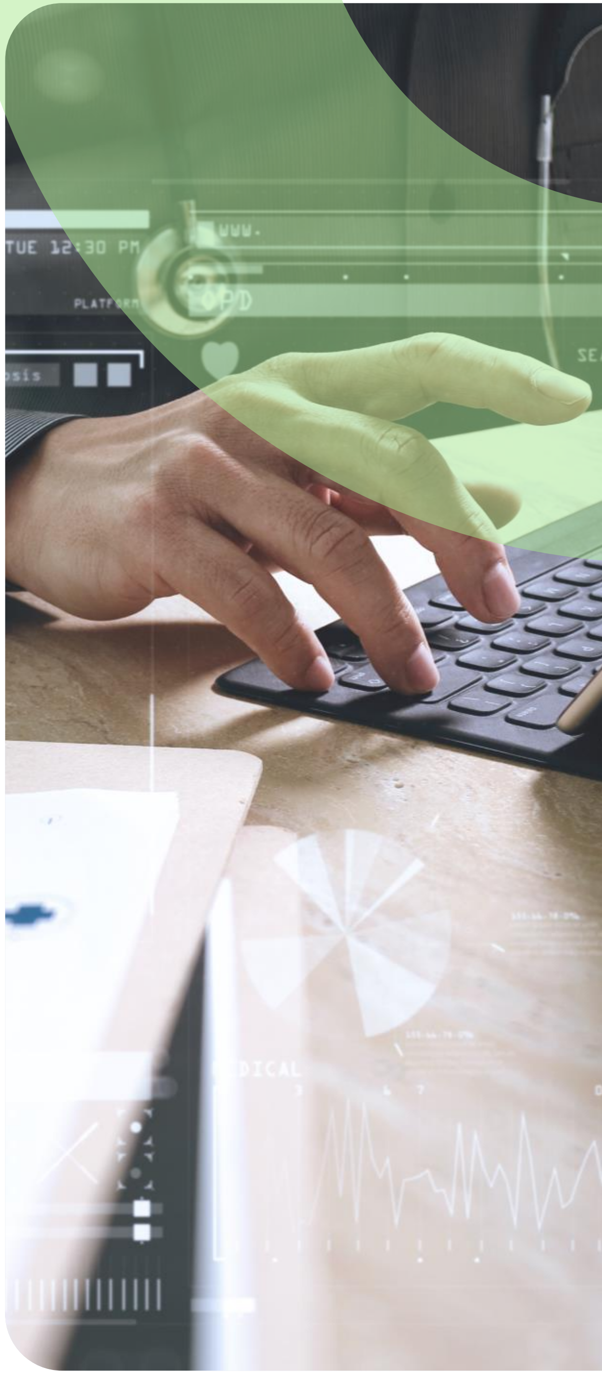
- More than one predictor variable
- $Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \ldots + \beta_n X_n + \varepsilon$

Residual standard error

- Estimate of standard deviation of ε
- Average deviation of response from true regression line
- Measure the lack of fit of the model

R^2 statistic (RSS)

- Measures amount of variability left unexplained by model
- 0 < RSS < 1
- Close to 0
  - Model does not explain variability in response well
- Close to 1
  - Model explains variability in response well



Model fit statistics in R

Summary(lm.fit)

Data frames

Tidyverse recap

- Collection of data analytics tools contained in R for transforming and visualising data
- Tibbles
  - Data frame with tweaks

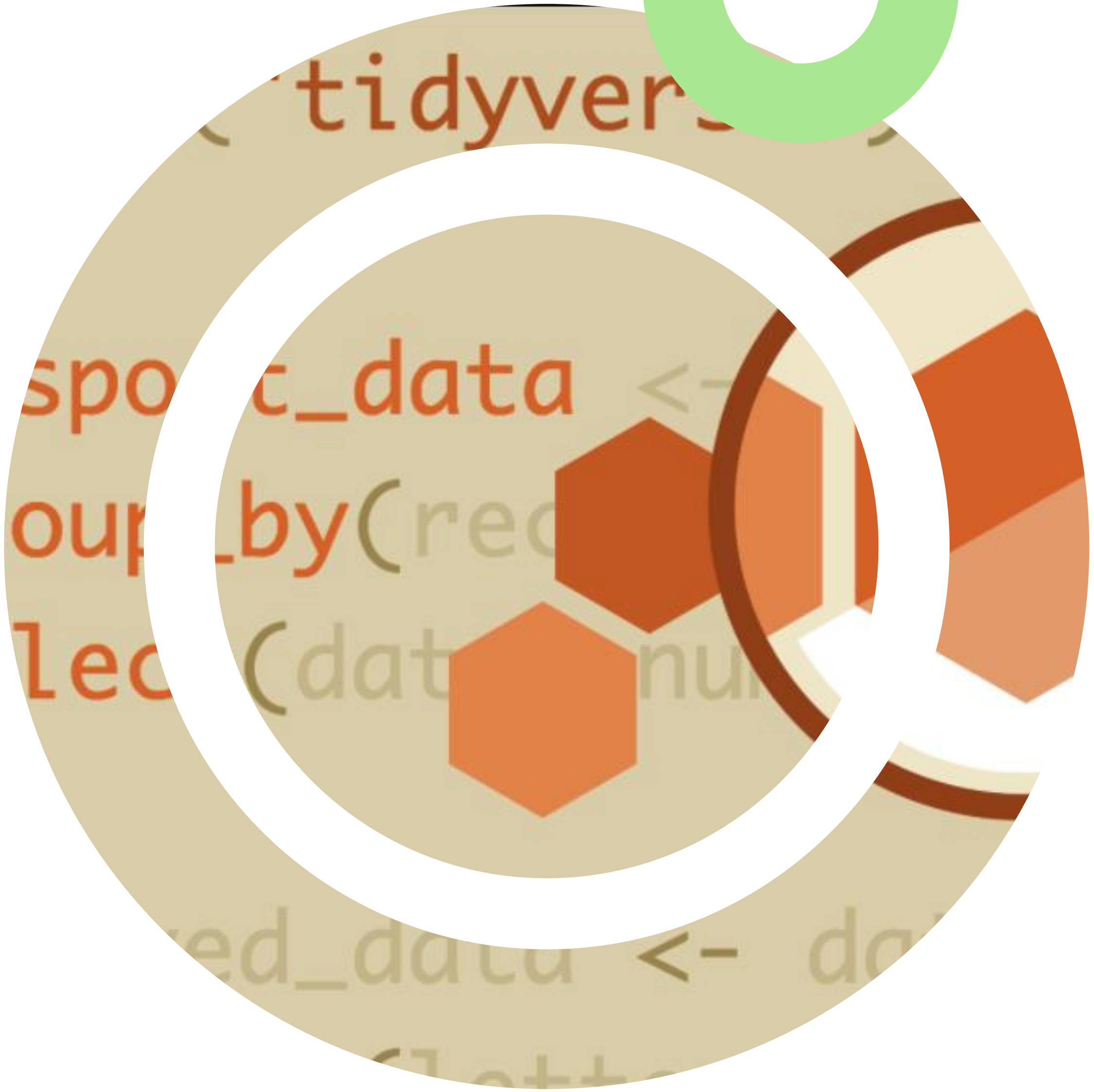


Image by: https://www.linkedin.com/learning/learning-the-r-tidyverse/what-is-the-tidyverse

More about tibbles

(package contained in tidyverse)

- Create tibble from data frame
  - As.tibble()
- Create new tibble
  - Tibble()
- Define a tibble row by row
  - Tribble()

```
Terminal × Jobs ×
cloud/project/
> mtcars
                     mpg cyl  disp  hp drat    wt  qsec vs am gear carb
Mazda RX4           21.0   6 160.0 110 3.90 2.620 16.46  0  1    4    4
Mazda RX4 Wag       21.0   6 160.0 110 3.90 2.875 17.02  0  1    4    4
Datsun 710          22.8   4 108.0  93 3.85 2.320 18.61  1  1    4    1
Hornet 4 Drive      21.4   6 258.0 110 3.08 3.215 19.44  1  0    3    1
Hornet Sportabout   18.7   8 360.0 175 3.15 3.440 17.02  0  0    3    2
Valiant             18.1   6 225.0 105 2.76 3.460 20.22  1  0    3    1
Duster 360          14.3   8 360.0 245 3.21 3.570 15.84  0  0    3    4
Merc 240D           24.4   4 146.7  62 3.69 3.190 20.00  1  0    4    2
Merc 230            22.8   4 140.8  95 3.92 3.150 22.90  1  0    4    2
Merc 280            19.2   6 167.6 123 3.92 3.440 18.30  1  0    4    4
Merc 280C           17.8   6 167.6 123 3.92 3.440 18.90  1  0    4    4
Merc 450SE          16.4   8 275.8 180 3.07 4.070 17.40  0  0    3    3
Merc 450SL          17.3   8 275.8 180 3.07 3.730 17.60  0  0    3    3
Merc 450SLC         15.2   8 275.8 180 3.07 3.780 18.00  0  0    3    3
Cadillac Fleetwood  10.4   8 472.0 205 2.93 5.250 17.98  0  0    3    4
Lincoln Continental 10.4   8 460.0 215 3.00 5.424 17.82  0  0    3    4
Chrysler Imperial   14.7   8 440.0 230 3.23 5.345 17.42  0  0    3    4
Fiat 128            32.4   4  78.7  66 4.08 2.200 19.47  1  1    4    1
Honda Civic         30.4   4  75.7  52 4.93 1.615 18.52  1  1    4    2
Toyota Corolla      33.9   4  71.1  65 4.22 1.835 19.90  1  1    4    1
Toyota Corona       21.5   4 120.1  97 3.70 2.465 20.01  1  0    3    1
Dodge Challenger    15.5   8 318.0 150 2.76 3.520 16.87  0  0    3    2
AMC Javelin         15.2   8 304.0 150 3.15 3.435 17.30  0  0    3    2
Camaro Z28          13.3   8 350.0 245 3.73 3.840 15.41  0  0    3    4
Pontiac Firebird    19.2   8 400.0 175 3.08 3.845 17.05  0  0    3    2
Fiat X1-9           27.3   4  79.0  66 4.08 1.935 18.90  1  1    4    1
Porsche 914-2       26.0   4 120.3  91 4.43 2.140 16.70  0  1    5    2
Lotus Europa        30.4   4  95.1 113 3.77 1.513 16.90  1  1    5    2
Ford Pantera L      15.8   8 351.0 264 4.22 3.170 14.50  0  1    5    4
Ferrari Dino        19.7   6 145.0 175 3.62 2.770 15.50  0  1    5    6
Maserati Bora       15.0   8 301.0 335 3.54 3.570 14.60  0  1    5    8
Volvo 142E          21.4   4 121.0 109 4.11 2.780 18.60  1  1    4    2
```

Tibble vs data frame

- Tibble have refined print method
  - Shows first 10 row, columns fit on screen, large data is easy to work with
- Subsetting
  - Stricter

Dates & times in R

Lubridate package in R

- Easier to handle dates and times in R
- Built by: H. Wickham & G. Grolemund
- Maintained by: V. Spinu

Shaw Academy

Current dates and times

- Name of current timezone: Sys.timezone()
- Current date: Sys.Date()
- Current date and time: Sys.time()

Lubridate package

- Current date and time: Now()

Converting dates to strings

- Convert strings to date format: as.Date(y, format = “%m/%d/%Y”)
- Lubridate package: ymd(x), mdy(x)

Challenge »

Create multiple linear regression models each with different predictor variables. Which model fits the data best and why?

#exploredata